

OMAR AHMED

AI & Software Engineer

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PROFESSIONAL SUMMARY

AI & Software Engineer with hands-on experience building responsive web applications and AI-powered systems. Skilled in Python, FastAPI, React.js, TypeScript, Machine Learning, Deep Learning, LLMs, RAG, and Agentic AI. Strong problem-solving background with experience integrating AI features into real software products.

EDUCATION

The Egyptian E-Learning University

Bachelor of Information Technology in Computer Science and IT

2022 – 2026

Alexandria, Egypt

EXPERIENCE

Digital Egypt Pioneers Initiative (DEPI)

Nov 2025 – July 2026

AI & Machine Learning Trainee

Alexandria, Egypt

- Built and evaluated ML/DL models in Python using modern AI engineering workflows and optimization techniques.
- Worked on data preprocessing, model training, performance evaluation, and practical AI implementation tasks.

Information Technology Institute (ITI)

Aug 2025 – Sep 2025

Generative AI Intern

Egypt

- Developed AI solutions using LLMs, Prompt Engineering, and deep learning frameworks for educational use cases.
- Applied structured prompting and AI workflow design to build user-focused intelligent outputs.

Digital Egypt Pioneers Initiative (DEPI) – New Horizons Alexandria

Oct 2024 – May 2025

Software Development Trainee / Front-End Developer Intern

Alexandria, Egypt

- Completed 10+ projects using HTML, CSS, JavaScript, React, TypeScript, and modern front-end workflows.
- Built responsive interfaces, reusable UI components, collaborative projects, and client-style presentations.

The Egyptian E-Learning University

Oct 2025 – Mars 2026

ICPC Assistant Coach

Alexandria, Egypt

- Mentored students in algorithms, data structures, and competitive programming problem-solving strategies.

PROJECTS

Nabdak – Heart Disease AI Platform

Graduation Project (A+) | Python, FastAPI, Machine Learning, LLMs, ECG, SHAP, LangChain, Groq API

- Led an 8-member graduation project and owned the AI track: ML risk prediction, LLM explanations, ECG analysis, and FastAPI AI service integration.
- Built an AI engine using 11 clinical features to classify heart disease risk, achieving about 95% accuracy on 238 patient profiles.
- Integrated ECG and LLM workflows to generate top-5 ECG findings, patient-friendly explanations, safety-aware recommendations, and downloadable reports.

Fitness Tracker Web App

React, TypeScript, Tailwind CSS, Chart.js

- Developed a responsive fitness dashboard for tracking calories burned, daily activity, workout progress, and friend performance.
- Built a mobile-friendly UI with reusable components, activity cards, progress charts, and performance visualizations.

AI-Powered Study Platform

Python, LangChain, LLMs, RAG, Prompt Engineering, AI Workflows

- Worked on the AI track of an educational platform that transforms uploaded study materials into interactive learning support.
- Implemented RAG-based tutoring logic, automatic quiz generation, personalized study plan generation, and LLM-powered learning workflows.
- Designed prompts and AI response flows to produce structured, useful, and student-friendly educational outputs.

CERTIFICATIONS & COURSEWORK

Certifications: Information Technology Specialist in Artificial Intelligence (Certiport) | Huawei HCIA-AI | React Frontend Web Developer | Building LLM Applications with Prompt Engineering

Relevant Coursework: Software Engineering | Machine Learning | Deep Learning | Pattern Recognition | Natural Language Processing

TECHNICAL SKILLS

Programming & Web: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, React.js, FastAPI

AI & Data Science: Machine Learning, Deep Learning, Generative AI, RAG, Agentic AI, PyTorch, TensorFlow, Scikit-learn, LangChain

Databases & Tools: PostgreSQL, Vector Databases, Supabase, Git, VS Code, Cisco Packet Tracer